Università degli studi di
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Reinforcement Learning

Final Project

Discount Factor as a Regularizer in Reinforcement Learning

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Abstract

This work aims to analyze the use of discount factor as a regularizer and formally presents a mathematical equivalence with L_2 regularization in value function estimation, showing how reducing the discount controls variance and improves stability in finite-data regimes.

1 Introduction

This report aims to define the role of the **discount factor** in Reinforcement Learning (RL), with a focus on its theoretical and practical implications for value function estimation. In standard RL formulations, the return is defined as the discounted sum of future rewards, where the discount factor $\gamma \in [0, 1)$ determines the relative importance of future outcomes. Amit et al. (2020) [1] analyze the role of the discount factor as a regularizer, a perspective that has received limited attention in the literature.

The authors focus on the theoretical point of view according to which solving a discounted RL problem with γ lower than the discount factor specified by the task, can be interpreted as solving a regularized version of the corresponding undiscounted objective. Under this interpretation, the discount factor controls the complexity and sensitivity of the value function, in a way analogous to L_2 regularization. Regularization techniques are commonly introduced to manage the bias-variance trade-off, and improve generalization. Therefore, by framing discounting as a regularization mechanism, the paper shows that smaller values of γ can reduce estimation variance and improve generalization performance in limited data regimes.

The main theoretical contribution of the paper is the research of a formal mathematical connection between discounting and an L_2-type regularization effect in value function estimation.

In this project, it has been analyzed the theoretical foundations of this reinterpretation. The work also reviews the experimental results presented in the paper and consider additional implementations inspired by examples discussed during the course of *Reinforcement Learning* [2].

Moreover, the contribution of the paper can be summarized through the following research questions: can reducing the discount factor improve generalization performance? How does the optimal discount factor depend on the amount of available data? How do properties of the dataset, such as state

distribution and mixing rate, affect the effectiveness of discount regularization? How does discount regularization compare to explicit L_2 regularization, both in tabular settings and under function approximation?

Theoretical Background In this section, the main theoretical concepts underlying the analysis presented in the paper are introduced.

A Reinforcement Learning problem is modeled as a Markov Decision Process (MDP), defined by a tuple $(S, A, P, R, \mu, \gamma_e)$.

The paper considers two settings: *policy evaluation* and *control*. In the policy evaluation setting, the agent is given a fixed policy π and aims to estimate its value function $V^{\pi}_{\gamma_e}(s)$. In the control setting, the goal is instead to find a policy π that maximizes the expected discounted return $\mathbb{E}_{\tau:\pi}\left[\sum_{t=0}^{\infty} \gamma_e^t r_t\right]$.

In particular, the paper distinguishes between: the evaluation discount factor γ_e, which defines the true objective of the problem and the guidance discount factor γ, which is used internally by the learning algorithm.

Different RL algorithms based on TD-style policy evaluation are employed to assess the contribution of the paper. For state-value estimation, the baseline method is TD(0), which performs one-step bootstrapped updates. To analyze the effect of longer horizons, the paper also considers n-step TD methods, with a focus on TD(3) in the Mujoco environment. In control settings, the objective shifts to estimating the action-value function Q^{π} rather than V^{π}. The extension of TD(0) to action-values is SARSA(0), and the authors also examine Expected SARSA(0). Finally, the analysis includes results obtained with the batch LSTD algorithm.

In Algorithm 1, as discussed during the course, a diminishing step-size $\alpha_i = \frac{1}{i}$ is assumed, which satisfies the classical Robbins–Monro and therefore guarantees convergence under mild assumptions. This choice also allows the theoretical results to focus on the statistical properties induced by the discount factor. In the case of function approximation, the algorithm relies on semi-gradient updates, which treat the bootstrapped target $\hat{V}_{\tilde{\theta}}(s')$ as fixed and compute gradients only w.r.t. the current value estimate.

2 The Methodological Approach

Algorithm 1 The analysis focuses on the Temporal-Difference family, and in particular on TD(0), which is used as the simplest setting to present the

main theoretical idea. Although the paper extends the results to SARSA, m-step TD and LSTD, TD(0) makes the equivalence more transparent.

The algorithm is considered in a batch setting. Access to a finite dataset $D = \{(s, a, r, s')\}$ is assumed, from which transitions are repeatedly sampled. The goal is not to study convergence speed, but to understand the effect of the discount factor under finite data.

Algorithm 1 is a generic form of batch semi-gradient TD(0) with two additional components: a global reward scaling factor ξ and an explicit regularization term $\Psi(s, \theta)$.

Algorithm 1 Generic Regularized Batch TD(0)

Hyper-parameters: $\gamma \in [0, \gamma_e]$, $\xi \in \mathbb{R}^+$ (global reward scaling), Ψ (regularization function)
Input: D
for $i = 0, 1, ..., N_{\text{iter}} - 1$ **do**
  Get uniformly random (s, a, r, s') from D
  $\theta_{i+1} := \theta_i + \alpha_i \Big(\xi r + \gamma \hat{V}_{\theta_i}(s') - \hat{V}_{\theta_i}(s)\Big) \nabla \hat{V}_{\theta_i}(s) - \alpha_i \nabla(\Psi)$.
end for

The term inside the parentheses is the TD error, potentially modified by reward scaling. As said in chapter 1.1, semi-gradient is used in standard TD(0) . In the standard non-regularized case, $\xi = 1$, $\Psi \equiv 0$, and $\gamma = \gamma_e$. The key idea of the paper is to distinguish between the evaluation discount γ_e and a guidance discount γ, with $\gamma \leq \gamma_e$, used during learning, and to analyze the theoretical implications of this choice.

Proposition 1 and its demonstration The core of the work is to formulate an equivalence between TD learning with reduced discount, and TD learning with an higher discount and an added regularization term.

Proposition 1 *Let θ_1, θ_2, ... parameters produced by Algorithm 1 using a discount factor $\gamma < \gamma_e$, with $\xi = 1$, $\Psi \equiv 0$, initial parameters θ_0 and learning rate α_i. The algorithm produces the same sequence of parameters θ_1, θ_2, ... if it is run with the discount factor $\gamma = \gamma_e$, but with added regularization function $\Psi(s, \theta) := \lambda \left(\hat{V}_\theta(s)\right)^2$, $\lambda := \frac{\gamma_e - \gamma}{2\gamma}$, reward scaling $\xi := \frac{\gamma_e}{\gamma}$, learning rate $\alpha'_i := \frac{\gamma}{\gamma_e}\alpha_i$ and the same initial parameters θ_0.*

The proposition is proved below, with each derivation step explicitly commented.

$$
\begin{aligned}
\theta_{i+1} &= \theta_i + \alpha_i \left(r + \gamma \hat{V}_{\theta_i}(s') - \hat{V}_{\theta_i}(s) \right) \nabla \hat{V}_{\theta_i}(s) && (1.1)\\
&= \phi_i + \alpha_i \left(r + \gamma \hat{V}_{\phi_i}(s') - \hat{V}_{\phi_i}(s) \right) \nabla \hat{V}_{\phi_i}(s) && (1.2)\\
&= \phi_i + \alpha_i \frac{\gamma}{\gamma_e} \left(\frac{\gamma_e}{\gamma} r + \gamma_e \hat{V}_{\phi_i}(s') - \frac{\gamma_e}{\gamma} \hat{V}_{\phi_i}(s) \right) \nabla \hat{V}_{\phi_i}(s) && (1.3)\\
&= \phi_i + \alpha_i' \left(\frac{\gamma_e}{\gamma} r + \gamma_e \hat{V}_{\phi_i}(s') - \hat{V}_{\phi_i}(s) + \hat{V}_{\phi_i}(s) - \frac{\gamma_e}{\gamma} \hat{V}_{\phi_i}(s) \right) \nabla \hat{V}_{\phi_i}(s) && (1.4)\\
&= \phi_i + \alpha_i' \left(\frac{\gamma_e}{\gamma} r + \gamma_e \hat{V}_{\phi_i}(s') - \hat{V}_{\phi_i}(s) \right) \nabla \hat{V}_{\phi_i}(s) - \alpha_i' \frac{\gamma_e - \gamma}{\gamma} \hat{V}_{\phi_i}(s) \nabla \hat{V}_{\phi_i}(s) && (1.5)\\
&= \phi_i + \alpha_i' \left(\frac{\gamma_e}{\gamma} r + \gamma_e \hat{V}_{\phi_i}(s') - \hat{V}_{\phi_i}(s) \right) \nabla \hat{V}_{\phi_i}(s) - \alpha_i' \nabla \left(\frac{\gamma_e - \gamma}{2\gamma} \left(\hat{V}_{\phi_i}(s) \right)^2 \right) && (1.6)\\
&= \phi_{i+1} && (1.7)
\end{aligned}
$$

The proof is based on inductive reasoning. Assume $\phi_i = \theta_i$; the goal is to show that $\phi_{i+1} = \theta_{i+1}$ as well. In (1.1) θ_{i+1} is defined as in Algorithm 1, with $\xi = 1$ and $\Psi \equiv 0$. The next step is to introduce ϕ_i using the inductive hypothesis and, as stated in the proposition, apply a rescaling of the hyper-parameters so that the update can be rewritten in terms of γ_e. In (1.3) this is done by scaling the learning rate as $\alpha_i' := \frac{\gamma}{\gamma_e}\alpha_i$ and scaling the reward by $\xi := \frac{\gamma_e}{\gamma}$. Equivalently, $\frac{\gamma}{\gamma_e}$ can be factored out from the TD error, absorbs it into the step-size, and compensates inside the brackets by multiplying the reward term by ξ and rewriting the discount term using γ_e.

In (1.4) the learning rate is rewritten using the shorthand α_i' for notational convenience, and within the brackets the same quantity $\hat{V}_{\phi_i}(s)$ has been added and subtracted. This is convenient because in (1.5) the expression is decomposed into two parts: the first one is a TD error with discount γ_e (and rescaled reward), while the second one is the residual term induced by using $\gamma < \gamma_e$.

In (1.6) the residual term is rewritten as the gradient of a quadratic penalty by using the chain rule $\hat{V}(s)\,\nabla\hat{V}(s) = \frac{1}{2}\nabla\left(\hat{V}(s)^2\right)$, which shows that it is exactly the gradient of a quadratic penalty on the value estimate. Therefore, (1.6) matches the update of Algorithm 1 with the rescaled hyper-parameters and with the activation regularizer $\Psi(s,\theta) = \lambda \hat{V}_\theta(s)^2$. Hence, the

right-hand side of (1.6) can be identified with ϕ_{i+1}, proving that $\phi_{i+1} = \theta_{i+1}$ and completing the inductive step.

In the same way, it is possible to express ϕ_{i+1} for SARSA(0) algorithm:

$$\phi_i + \alpha_i' \left(\frac{\gamma_e}{\gamma} r + \gamma_e \hat{V}_{\phi_i}(s') - \hat{Q}_{\phi_i}(s,a) \right) \nabla \hat{Q}_{\phi_i}(s,a) - \alpha_i' \nabla \left(\frac{\gamma_e - \gamma}{2\gamma} \left(\hat{Q}_{\phi_i}(s,a) \right)^2 \right)$$

Below it is provided its pseudo code:

Algorithm 3 Batch Expected SARSA(0) with activation regularization

Hyper-parameters: $\lambda \in \mathbb{R}^+$ (regularization factor), $\xi \in \mathbb{R}^+$ (global reward scaling)
Input: D, π
for $i = 0, 1, ..., N_{\text{iter}} - 1$ **do**
  Get uniformly random (s, a, r, s') from D
  $\hat{V}_i(s') := \sum_{a' \in \mathcal{A}} \pi(a'|s') \hat{Q}_{\phi_i}(s', a')$
  $\phi_{i+1} := \phi_i + \alpha_i' [\xi r + \gamma_e \hat{V}_i(s') - \hat{Q}_{\phi_i}(s,a)] \nabla \hat{Q}_{\phi_i}(s,a) - \alpha_i' \nabla \left(\lambda \left(\hat{Q}_{\phi_i}(s,a) \right)^2 \right)$.
end for

Activation Regularization With the introduction of the regularization term, referred to as *activation regularization*, an additional inductive bias is imposed on the value estimates. Since the penalty acts directly on the magnitude of the estimated value function, it discourages large value predictions and can improve generalization by reducing the impact of spurious approximation errors. From Proposition 1, the regularization coefficient is $\lambda = \frac{\gamma_e - \gamma}{2\gamma}$. This expression makes explicit the connection between discount reduction and regularization strength: when the guidance discount γ is decreased (with $\gamma < \gamma_e$), the corresponding regularization coefficient λ increases.

From an optimization perspective, running TD(0) with a reduced discount $\gamma < \gamma_e$ is equivalent to minimizing

$$\mathbb{E}_{(s,a,r,s') \sim D} \left[\left(\xi r + \gamma_e \hat{V}_{\tilde{\theta}}(s') - \hat{V}_\theta(s) \right)^2 + \lambda \left(\hat{V}_\theta(s) \right)^2 \right]$$

where the additional quadratic term corresponds exactly to the activation regularizer.

This equivalence becomes more explicit under structural assumptions on the value function. Consider a linear approximation $\hat{V}_\theta(s) = \phi(s)^\top \theta$, with fixed features $\phi(s) \in \mathbb{R}^k$ and parameters $\theta \in \mathbb{R}^k$. Assume that the features are orthogonal under the sampling distribution, that is $\mathbb{E}_{s \sim D} \left[\phi(s) \phi(s)^\top \right] = \Lambda$,

with Λ diagonal. Under this assumption, the activation regularization term becomes

$$\lambda\, \mathbb{E}_{s\sim D}\left[\left(\hat{V}_\theta(s)\right)^2\right] = \lambda\, \mathbb{E}_{s\sim D}\left[(\theta^\top \phi(s))^2\right] = \lambda\, \theta^\top \Lambda \theta,$$

which corresponds to a weighted ℓ_2 penalty on the parameters, where the weights are given by the diagonal entries of Λ.

The tabular representation is a special case of this setting. If $\phi(s) = e_s$ (standard basis vectors), then the second-moment matrix is diagonal by construction, and the activation penalty reduces to a per-state quadratic penalty on the corresponding parameters.

In the further case where the features are orthonormal, so that $\Lambda = I$, the activation regularization reduces to the standard ridge penalty $\lambda\|\theta\|_2^2$. In this scenario, a penalty on the outputs $\hat{V}(s)$ becomes indistinguishable from an ℓ_2 shrinkage on the parameters. However, it is important to note that, to obtain an exact equivalence at the level of update dynamics, one must also apply the compensations introduced in Proposition 1, namely the reward rescaling $\xi = \gamma_e/\gamma$ and the step-size rescaling $\alpha_i' = (\gamma/\gamma_e)\alpha_i$.

Uniformity of the distribution The expression $\lambda\mathbb{E}_{(s)\sim D}\left(\hat{V}_\theta(s)\right)^2$ allows to discuss the distribution over the observed states. In the tabular form the activation regularization can be expressed as $\lambda\mathbb{E}_{(s)\sim D}\theta^2$, meaning that if a state is less visited, or not visited at all, the regularization is low, or not regularized. This suggests that activation regularization is more effective when the empirical distribution is closer to uniform and this property will be taken into account in the experiments, i.e. in the tabular case by sampling from an uniform policy. To quantify how close the empirical state-visitation distribution is to uniform, the *total variation distance* [3] is computed between the empirical distribution d_D induced by the dataset D and the uniform distribution u over $\mathcal{S}$, i.e. $u(s) = 1/|\mathcal{S}|$. In the discrete case, the total variation distance is

$$\mathrm{TV}(d_D, u) = \frac{1}{2}\sum_{s\in\mathcal{S}} |d_D(s) - u(s)| = \frac{1}{2}\sum_{s\in\mathcal{S}} \left| d_D(s) - \frac{1}{|\mathcal{S}|}\right|.$$

Mixing time analysis Another factor that affects the effectiveness of activation regularization and, equivalently, using a smaller discount factor during

learning, is the *mixing time* [4]. For a Markov chain induced by a fixed policy, the mixing time describes how quickly the state distribution approaches the stationary distribution. A common way to relate mixing properties to the transition dynamics is through the *spectral gap* of the transition matrix: slower mixing corresponds to a smaller spectral gap, and the mixing time can be estimated (up to constants) by the inverse of this gap.

$$\tau_{\text{mix}} \propto \frac{1}{\text{gap}(P)}, \qquad \text{gap}(P) = 1 - |\lambda_2(P)|$$

where $\lambda_2(P)$ denotes the second largest eigenvalue (in magnitude) of the transition matrix P. Intuitively, a large τ_{mix} means *slow mixing*: trajectories tend to remain for long periods in restricted regions of the state space. This produces samples that are more correlated and less representative of the full environment. This matters for activation regularization because the penalty term $\lambda\, \mathbb{E}_{s\sim D}\big[\hat{V}_\theta(s)\big]^2$ is weighted by the empirical state-visitation distribution induced by the dataset D. As a result, frequently visited states are strongly regularized, while rarely visited (or unseen) states receive little or no regularization. When mixing is slow, coverage becomes uneven and estimation variance increases. In this setting, stronger regularization is typically beneficial.

3 Results and Evaluation

The authors ran empirical demonstrations to investigate their theories. Due to computational limitations, some parameters were reduced. Different scenarios were analyzed to understand effects of size, uniformity of distribution and time mixing.

Tabular experiments The effects of discount regularization were investigated in a GridWorld experiment. Here, the state space is a 4x4 grid along which the agent moves. Goal states, reward means, and transition probabilities are generated randomly. LSTD and batch-TD are used for policy evaluation while SARSA for policy optimization.

The first experiment is run on batch policy evaluation setting, for a fixed uniform policy. The evaluation metric is $\left\|\hat{V} - V^{\pi}_{\gamma_e}\right\|_2$, where $V^{\pi}_{\gamma_e}$ is evaluated with $\gamma_e = 0.99$. In this experiment, the number of MDP instances was reduced to 50 and to only two trajectories for episode (4 and 16).

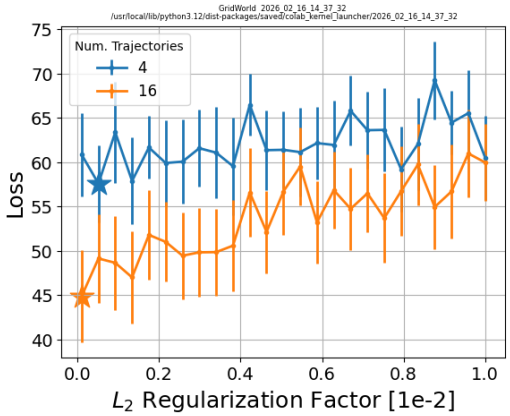
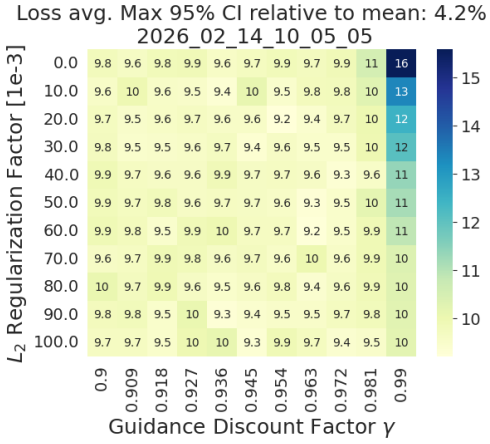


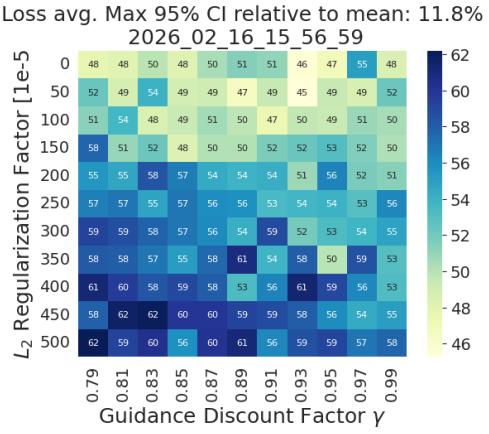
Figure 1: Tabular experiment, policy evaluation with SARSA. Use of L_2 regularization over different trajectories.Error bars represent 95% confidence interval.

The plot (Fig.1) shows that with higher trajectories the learned policy is closer to the optimal, intuitively, as more data are available. With 4 trajectories, L_2 regularization helps achieve the minimum while with more data is less influential. As stated above, L_2 appears to be more beneficial with fewer data.

The next experiment aims to evaluate the effect of L_2 regularization and discount factor γ on policy evaluation. The algorithm used is LSTD with 1000 repetitions.



(a) Tabular experiments: loss values for different combinations of L_2 and γ. Policy evaluation with LSTD over 1000 repetitions.



(b) Tabular experiments: effects of L_2 and γ regularization over different trajectories. Policy optimization with SARSA over 50 repetitions.

The heatmap (Fig.2a) shows variation in loss values due to different combinations of L_2 and γ. Lower values of loss are found within smaller values

of regularization, and discount factor around 0.95. As stated above, when γ gets closer to γ_e there is more instability, showing that more regularization or a lower discount factor is needed.

Next, the objective is to evaluate policy optimization over different combinations of L_2 and γ. Moreover, quantify how is the optimal policy learned using SARSA in a tabular MDP.

These results (Fig.2b) are more unstable (11% off the mean) as it was run only over 50 repetitions and there is more exploration $\epsilon - greedy$. The loss is higher when γ is higher, so closer to γ_e. A proof that lower discount factors are acting as an implicit regularization.

The next point is to evaluate the influence of non-uniform distributions. Here, a batch setting is considered, where state-actions tuples are extracted independently from fixed distributions, while rewards, next actions, and states are drawn accordingly to the policy and environment. Distributions are generated via rejection sampling, and TV is the total variation distance from a uniform distribution.

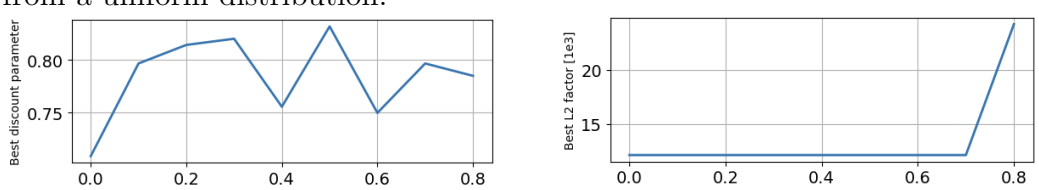


Figure 3: On x-axis total variation from uniform (normalized). Tabular experiments: effects of different combinations of L_2 and γ in variation from uniform distributions.

The plot (Fig.3) shows how the regularization parameter varies with changes in the uniformity of the data. A lower discount factor is needed when the distribution is close to being uniform, while it fluctuates when the distribution moves away from uniformity.

The empirical results are in line with the theoretical interpretations of the paper. To summarize, the following plot (Fig.4) shows the instability introduced with non-uniform distribution in policy evaluation with MRP.

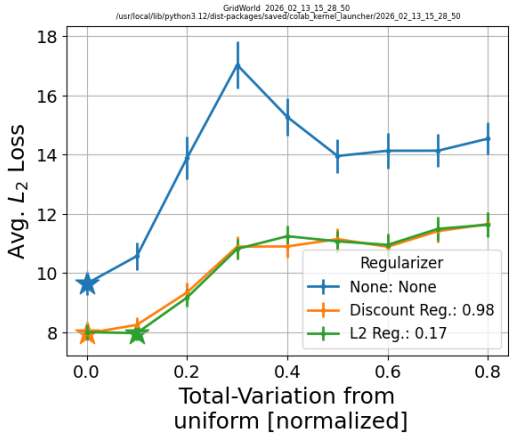
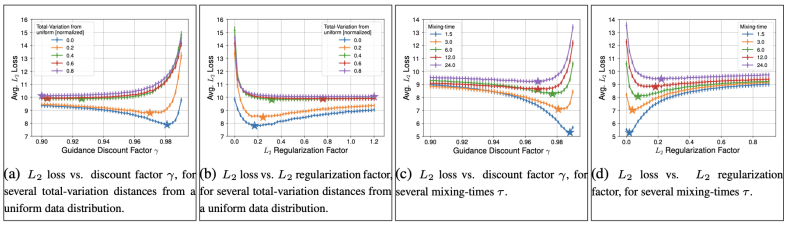


Figure 4: Tabular settings, policy evaluation with MRP. Error bars represent 95% confidence interval.

As TV increases, the estimate of the value is more complicated and more regularization is required. Both L_2 regularization and the implicit version given by reducing γ have a similar effect, especially when the data is unbalanced, confirming the theories of the original paper.
At this point, time mixing was considered in the analysis. In a Markovian setting, it describes the convergence time from a state to a stationary distribution. To apply this concept to the trajectories (2 of length 50), the authors augmented the transition probabilities and estimated the values in a batch policy evaluation setting with LSTD. For each experiment, the authors randomly choose an MDP, from which they derived the Markov process induced by a uniform policy and applied the time augmenting procedure. Due to computational and software versions limitation, this part was not reproduced. The graph from the original paper is displayed below:



(a) L_2 loss vs. discount factor γ, for several total-variation distances from a uniform data distribution.

(b) L_2 loss vs. L_2 regularization factor, for several total-variation distances from a uniform data distribution.

(c) L_2 loss vs. discount factor γ, for several mixing-times τ.

(d) L_2 loss vs. L_2 regularization factor, for several mixing-times τ.

Figure 5: Tabular Experiments - effect of data properties. Loss vs. regularization factor for different regularizers, averaged over 1000 MDP instances. All results are with the LSTD algorithm. The star shapes mark the minimum of the curve. Error bars represent 95% confidence interval.

The (Fig.5) shows that both regularization are more effective in the slow mixing regime. Intuitively, when less data is available, variance increases and

consequently the need of regularization.

Deep RL Experiments The authors wanted to investigate the effects of regularization on continuous control with function approximation settings. The experiments employs Mujoco environment [5]. Due to computational limitations, the time-steps were reduce to 25,000 and experiments were not repeated for 20 different initial random seeds as was done in the original work. Learning rate, algorithm (a variant of expected SARSA), batch size (256) and hyperparameters were kept the same of the original paper.

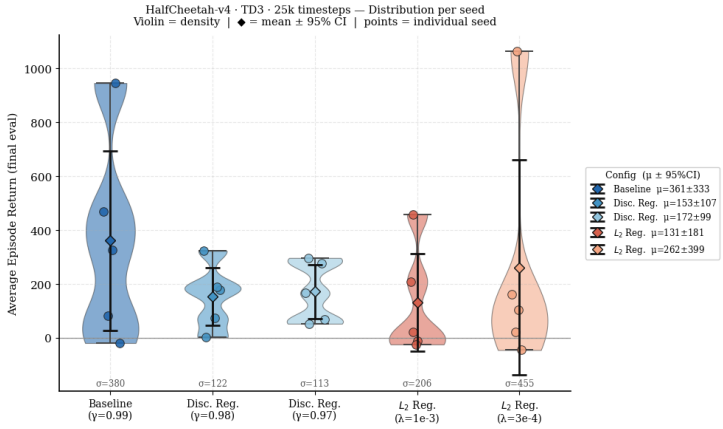


Figure 6: Agents rewards under different configurations.

Results are summarized in Fig.6. There are five configurations in sweep TD3.

- *Baseline* (γ=0.99, L_2=0): High bootstrapping and no regularization.
- $DiscReg_{0.98}$ (γ=0.98, L_2=0): For the proposition of the paper this means adding λ=0.01/(2*0.98) = 0.005 as a penalty on $||\hat{V}(s)||^2$. The critic is more conservative.
- $DiscReg_{0.97}$ (γ=0.97, L_2=0), lower discount factor.
- $L2_{1e-3}$ (γ=0.99, L_2=1e-3): Explicit regularization on parameters instead of implicit regularization through γ.
- $L2_{3e-4}$ (γ=0.99, L_2=3e-4): Similar to DiscReg. One seed seems to have found optimal solution. However, with too limited data L_2 is not able to regularize systematically.

Results of sweep TD3 on HalfCheetah-v4 (with 25k timesteps, 5 seeds) show high variance among all configurations, consistent with the limited data. The baseline has higher mean, but it is dominated by one outlier seed making the comparison not conclusive. Configurations with lower discount show a lower standard deviation, suggesting that discount regularization might stabilize even with fewer seeds. L_2 configurations are instable between seeds. Further conclusions cannot be drawn, as more seeds are needed.

Application on Windy Gridworld These theories were further evaluated on one example taken from the exercises of the course *Reinforcement Learning* [2]. Regularization was applied to Windy Gridworld.

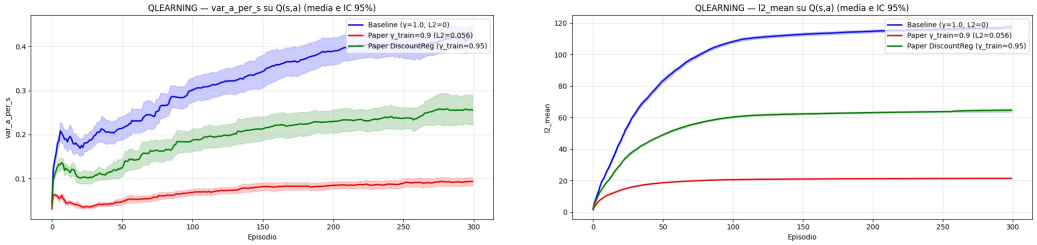


Figure 7: Tabular exercise: Windy gridworld exercise with three settings. Variance and L_2 norm are shown.

In Figure 7 the quantity *var a per s* $= \mathbb{E}_s[\mathrm{Var}_a(Q(s,a))]$ on the left shows that the baseline configuration $\gamma = 1$ produces a significantly larger dispersion of action-values within each state, while both discount-based regularization schemes substantially reduce this variance. This indicates that smaller guidance discounts compress the Q-function, consistently with the quadratic penalty interpretation. The plot on the right shows the ℓ_2-mean of Q(s,a) directly reflects the theoretical prediction of Proposition 1: reducing γ leads to a strong shrinkage of the value function norm, with the baseline reaching the largest magnitude and the more aggressively regularized setting the smallest. In the Source code provided there are two more plots: one regards the greedy final evaluation and shows that, although average performance is comparable, the baseline exhibits much larger variability across runs, whereas discount-regularized versions produce more stable policies. Overall, these results empirically support the interpretation of the discount factor as an implicit regularizer that controls both the magnitude and the stability of the learned value function. The last one depicts the evolution of the training returns suggesting that all methods converge to similar performance levels,

but the regularized variants exhibit slightly more stable dynamics during the early learning phase.

4 Conclusions

This project investigated the interpretation of the discount factor as a regularization mechanism, focusing on its role in value estimation and policy optimization under limited data.

In the tabular experiments, reducing the discount factor improved stability in low-data regimes. When the number of trajectories was small, both explicit L_2 regularization and discount reduction led to lower estimation error.

The mixing time analysis demonstrated that, in environments characterized by slower mixing and higher sample correlation, stronger regularization improved performance. This aligns with the hypothesis that reducing γ improves conditioning and mitigates variance growth in settings with small spectral gap.

In the deep RL experiments, due to computational constraints, conclusions are less robust. The limited number of seeds and reduced training steps introduce high variance. However, configurations with lower discount factors exhibited lower standard deviation across seeds, suggesting a stabilizing effect. In contrast, explicit L_2 regularization showed higher sensitivity to initialization and did not consistently outperform discount-based regularization. Even if these results are not conclusive, they are aligned with the theoretical hypothesis.

Several limitations of the experimental setup were found. The number of repetitions in some tabular experiments was relatively small, which increases variance in the metrics. In the deep RL setting, the reduced number of seeds and timesteps limits statistical reliability and makes it difficult to draw strong conclusions.

In conclusion, the empirical evidence supports the view that the discount factor can act as a regularization mechanism, particularly in low-data and poorly conditioned regimes.

References

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